IDLE Shell 3.13.17

File Edit Shell Debug Options Window Help

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Python 3.13.17 (tags/v3.13.17:64e080d, Oct 21 2026, 14:08:53) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
== RESTART: C:/Users/josth/AppData/Local/Programs/Python/Python313/task_11.py ==
TASK 11: FINAL DESIGN RECOMMENDATION
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Litres: 10
Fat: 3.5
SNF: 8.5
Milk Rate: Rs. 30.00
Payment: Rs. 300.00

Litres: 8.5
Fat: 4.2
SNF: 8.7
Milk Rate: Rs. 33.40
Payment: Rs. 283.90

Litres: 5
Fat: 3.0
SNF: 8.0
Milk Rate: Rs. 26.50
Payment: Rs. 132.50

ROUNDING RULE
Original value: 10.125
ROUND_HALF_UP: 10.13

RATE REVISION
Old base rate: Rs. 30.00
New base rate: Rs. 31.00
New rate for reference milk: 31.00

JSON RATE CARD COMPARISON
Rate card loaded from: rate_card.json
JSON file size: 123 bytes
File loading time: 188.3 microseconds

FINAL RECOMMENDATION
1. Use Decimal for accurate monetary calculations.
2. Use ROUND_HALF_UP for rounding to two decimals.
3. Store rates in a JSON file for easier revisions.
4. Validate inputs and test normal and invalid values.
5. Keep rate changes approved and recorded.
6. Use a database and audit logs at district scale.
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LIMITATIONS
The program requires valid input and an approved rate card.
It does not support multiple users or database storage.
District deployment needs security, backups and testing.

All experiments completed.
>>>
```